

COURSE DESCRIPTION			
Program	Chemical Engineering		
Course Title	Industrial Safety Lab		
Course Code	2078	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

The Industrial Safety Lab provides students with hands-on exposure to essential safety principles and practices in chemical and allied industries. The lab focuses on the correct use of PPE and safety signage, fire extinguisher selection and operation, mock drills, evacuation procedures and basic first aid. The lab also emphasises safe handling of chemicals and glassware through the use of safety data sheets, proper storage, labelling, and transfer methods. Students are introduced to foundational concepts for higher-risk operations such as lockout-tagout, energy isolation, confined-space entry and gas testing. They also learn to document and analyse unwanted events using simple incident reporting formats and root cause analysis tools so that they can develop a strong, practice-oriented safety mindset applicable to laboratories and process plants.

Course Objectives

1	To identify and apply basic personal safety and emergency safety practices.
2	To demonstrate safe chemical and laboratory work practices.
3	To describe and practice fundamental controls for permit-controlled operations and isolation.
4	To record and analyse safety incidents for continual improvement.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic understanding of force, pressure, heat, and electricity	2002B	Engineering Physics for Applied Electrical Technology and Computing	I
Basic understanding of acids, bases, reactions, and flammable substances	2003A	Chemistry for Engineering Practices	I
Basic knowledge of hazards in laboratories	2009B	Engineering Workshop Practice	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Comprehend essential industrial and laboratory safety requirements.	Apply
CO2	Apply standard safety procedures.	Apply
CO3	Comprehend permit systems.	Apply
CO4	Document and analyse incidents to enhance safety.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	3	2	3	3	2	2
CO2	-	-	-	2	-	3	2	3	3	2	2
CO3	-	-	-	3	-	3	2	3	3	2	2
CO4	-	-	-	3	-	3	2	3	3	2	2
Total	-	-	-	8	-	12	8	12	12	8	8
Correlation Level	-	-	-	2.33	-	3	2	3	3	2	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Personal Protective Equipment (PPE) selection and fitting drill.	Select, inspect, and use appropriate PPE (Safety Helmet, Safety Goggles, Face shield, Safety Shoes, Chemical-resistant Gloves, Ear Plugs/Ear Muffs, and Safety Apron/Lab Coat) based on identified workplace hazards.	CO1	Apply	3
2	Hazard Identification	Systematically identify and classify hazards and suggest basic control measures to minimise associated risks. (HAZID Walkthrough - Identify 10 hazards in the lab - slips, chemicals, electrical, fire, etc.)	CO1	Apply	3
3	Fire Extinguisher	Select and use appropriate Fire Extinguishers for different types of fire. (PASS technique practice.)	CO1	Apply	3
4	Fire hydrants	Demonstrate fire hydrant and valve operation. Hose reel deployment and nozzle handling.	CO1	Apply	3
5	Mock Drill	Demonstrate the evacuation drill using the nearest, safest route. (Fire alarm, gas leak response, and assembly verification)	CO1	Apply	3
Module II					
6	MSDS	Decode the Material Safety Data Sheets and select the suitable PPE for safe handling of the chemical. Prepare a hazard rating comparison across the chemicals used in the laboratory	CO2	Apply	3
7	Safe Laboratory Practices	Demonstrate Glassware Handling & Safe Lab Practices for all the labs included in the curriculum	CO2	Apply	3
8	Emergency Response	Demonstrate the Neutralization of an acid/alkali spill. Operate and demonstrate the proper use of the Eyewash and Shower. Perform basic first aid measures for minor chemical exposure, burns, or injuries under supervision.	CO2	Apply	3
	Series Exam - I		CO2	Apply	3
Module III					
9	LOTO	Prepare a Lockout-Tagout (LOTO) permit and simulate LOTO on lab equipment	CO3	Apply	3
10	Confined Space entry	Prepare a Confined-Space entry permit.	CO3	Apply	3
Module IV					
11	Incident Reporting	Demonstrate Near-Miss Incident Reporting	CO4	Apply	3
12	Root Cause Analysis, Corrective and Preventive Action	Apply the 5 Why analysis technique to identify the root cause of a given incident and propose appropriate corrective and preventive actions.	CO4	Apply	3
	Series Exam - II		CO4	Apply	3
	Open Ended Experiment		CO4	Apply	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
 Integrate multiple concepts from the syllabus into a combined application.
 Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
 Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Study PPE types for chemical handling and create a selection checklist. Prepare the lab record for the PPE selection and fitting drill	1.5
2	Watch recorded video and spot hidden hazards in lab safety videos. Prepare the lab record and log for Hazard identification with photos	1.5
3	Prepare flashcards to match fire classes (A,B,C,D,K) to extinguisher types. Prepare the lab record with the fire class-extinguisher matching table	1.5
4	Watch the recorded video on the hydrant, fire monitor usage, and prepare a summary. Prepare the lab record.	1.5
Module II		
5	Prepare an escape route from any one laboratory to the assembly point. Prepare the lab record.	1.5
6	Compare MSDS for 5 lab chemicals downloaded from the manufacturer's website; note PPE differences. Prepare the lab record.	1.5
7	Watch the recorded video on glassware safety rules, and prepare a list of common mistakes. Prepare the lab record.	1.5
8	Study and tabulate neutralisation agents for 5 common lab spills. Prepare the lab record.	1.5
Module III		
9	Review Weeks 1-8 records; prepare viva questions for the lab exam.	1.5
10	Draw the LOTO sequence flowchart for pump maintenance. Prepare the lab record.	1.5
11	Watch the recorded video on confined space entry. List different gas detector types. Prepare the lab record.	1.5
12	Write a fictional near-miss report from a lab observation. Prepare the lab record.	1.5
Module IV		
13	Practice 5-Whys on a personal daily mishap example. Prepare the lab record.	1.5
14	Open-ended experiment: Research advanced topics and prepare for open-ended experiments.	1.5
15	Review lab records and prepare Viva questions for the lab exam.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Fundamentals of industrial Safety and health	Dr. K. U. Mistry	Siddharth Prakashan
2	Safety and hazard Management in chemical Industries	Prof .M. N Vyas	Atlantic

REFERENCE			
Sl. No.	Title	Author	Publication
1	Chemical Process Safety: Fundamentals with Applications	Daniel A. Crowl & Joseph F. Louvar	Pearson

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	www.nptel.ac.in	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
31	Safety Tongs / Forceps	Stainless steel	6
32	LOTO Training Kit	Electrical + valve	2 kits
33	Safety Padlocks	Keyed-different	12
34	LOTO Tags	Reusable	30
35	Safety lockout Hasps	Steel	6
36	Dummy Switch / Valve	For practice	2
37	Tripod with Dummy Harness	Training model	1
38	Warning Barricade / Cone	Safety colour	6
39	Confined Space Signage	Laminated	4
40	White Board	With markers	
41	Notice Board	Pin-up type	
42	First Aid Box	IS / ISO compliant	6
43	Burn Dressing Kit	Demo type	2

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10

Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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